Exponential & Logarithmic Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Negative exponents

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(5)$.

Final Answer: 23. Substitute $x = 5$: $f(5) = 4(5) + 3 = 23$.

2 [Foundational] • Target: Negative exponents

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

3 [Foundational] • Target: The laws of exponents and scientific notation

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(2)$.

Final Answer: 14. Substitute $x = 2$: $f(2) = 5(2) + 4 = 14$.

4 [Foundational] • Target: The laws of exponents and scientific notation

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Final Answer: $x = 1$ and $x = 6$. Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

5 [Foundational] • Target: The laws of exponents and scientific notation

A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

6 [Foundational] • Target: The laws of exponents and scientific notation

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(3)$.

Final Answer: 19. Substitute $x = 3$: $f(3) = 5(3) + 4 = 19$.

7 [Foundational] • Target: The laws of exponents and scientific notation

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

Final Answer: $x = 2$ and $x = 7$. Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: The laws of exponents and scientific notation

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(4)$.

Final Answer: 24. Substitute $x = 4$: $f(4) = 5(4) + 4 = 24$.

9 [Intermediate] • Target: The laws of exponents and scientific notation

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

Final Answer: $x = 3$ and $x = 8$. Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

10 [Intermediate] • Target: The laws of exponents and scientific notation

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(5)$.

Final Answer: 29. Substitute $x = 5$: $f(5) = 5(5) + 4 = 29$.

11 [Intermediate] • Target: The laws of exponents and scientific notation

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Final Answer: $x = 4$ and $x = 9$. Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

12 [Intermediate] • Target: Patterns with rational and irrational numbers

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(2)$.

Final Answer: 17. Substitute $x = 2$: $f(2) = 6(2) + 5 = 17$.

13 [Intermediate] • Target: Patterns with rational and irrational numbers

A data set recorded for patterns with rational and irrational numbers consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 24.5. The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

14 [Intermediate] • Target: Patterns with rational and irrational numbers

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(3)$.

Final Answer: 23. Substitute $x = 3$: $f(3) = 6(3) + 5 = 23$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Patterns with rational and irrational numbers • Cross-Lesson: 1.01 Linear Equations & Lines

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(4)$.

Final Answer: 29. Substitute $x = 4$: $f(4) = 6(4) + 5 = 29$.

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16 [Challenging] • Target: Patterns with rational and irrational numbers • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(5)$.

Final Answer: 35. Substitute $x = 5$: $f(5) = 6(5) + 5 = 35$.

17 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 11.01 Statistics & Data Analysis

For the model $f(x) = 2x + 6$ in simplifying radical expressions, evaluate $f(2)$.

Final Answer: 10. Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

18 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 2.01 Quadratic Functions

A data set recorded for simplifying radical expressions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 16.5. The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

19 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 2x + 6$ in simplifying radical expressions, evaluate $f(3)$.

Final Answer: 12. Substitute $x = 3$: $f(3) = 2(3) + 6 = 12$.

20 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 2x + 6$ in simplifying radical expressions, evaluate $f(4)$.

Final Answer: 14. Substitute $x = 4$: $f(4) = 2(4) + 6 = 14$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiplication properties of exponents

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(2)$.

a. 12 ✓

b. 14

c. 4

d. 16

Substitute $x = 2$: $f(2) = 2(2) + 8 = 12$.

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22 [Foundational] • Target: Multiplication properties of exponents

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Multiplication properties of exponents

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(5)$.

a. 18 ✓

b. 20

c. 10

d. 22

Substitute $x = 5$: $f(5) = 2(5) + 8 = 18$.

24 [Foundational] • Target: Division properties of exponents and the zero exponent

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(2)$.

a. 15 ✓

b. 18

c. 6

d. 19

Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

25 [Foundational] • Target: Division properties of exponents and the zero exponent

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Division properties of exponents and the zero exponent

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(5)$.

a. 24 ✓

b. 27

c. 15

d. 28

Substitute $x = 5$: $f(5) = 3(5) + 9 = 24$.

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27 [Foundational] • Target: Negative exponents

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(2)$.

a. 11 ✓

b. 15

c. 8

d. 15

Substitute $x = 2$: $f(2) = 4(2) + 3 = 11$.

28 [Foundational] • Target: Negative exponents

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiplication properties of exponents

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Multiplication properties of exponents

A data set recorded for multiplication properties of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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31 [Intermediate] • Target: Multiplication properties of exponents

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(3)$.

a. 14 ✓

b. 16

c. 6

d. 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

32 [Intermediate] • Target: Multiplication properties of exponents

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(4)$.

a. 16 ✓

b. 18

c. 8

d. 20

Substitute $x = 4$: $f(4) = 2(4) + 8 = 16$.

33 [Intermediate] • Target: Multiplication properties of exponents

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

34 [Intermediate] • Target: Multiplication properties of exponents

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

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35 [Intermediate] • Target: Division properties of exponents and the zero exponent

A data set recorded for division properties of exponents and the zero exponent consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

36 [Intermediate] • Target: Division properties of exponents and the zero exponent

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(3)$.

a. 18 ✓

b. 21

c. 9

d. 22

Substitute $x = 3$: $f(3) = 3(3) + 9 = 18$.

37 [Intermediate] • Target: Division properties of exponents and the zero exponent

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(4)$.

a. 21 ✓

b. 24

c. 12

d. 25

Substitute $x = 4$: $f(4) = 3(4) + 9 = 21$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Division properties of exponents and the zero exponent • Cross-Lesson: 2.01 Quadratic Functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

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39 [Challenging] • Target: Division properties of exponents and the zero exponent • Cross-Lesson: 3.01 Exponents & Polynomials

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

40 [Challenging] • Target: Negative exponents • Cross-Lesson: 4.01 Functions & Transformations

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

41 [Challenging] • Target: Negative exponents • Cross-Lesson: 6.01 Systems & Inequalities

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(3)$.

a. 15 ✓

b. 19

c. 12

d. 19

Substitute $x = 3$: $f(3) = 4(3) + 3 = 15$.

42 [Challenging] • Target: Negative exponents • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(4)$.

a. 19 ✓

b. 23

c. 16

d. 23

Substitute $x = 4$: $f(4) = 4(4) + 3 = 19$.

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43 [Challenging] • Target: Negative exponents • Cross-Lesson: 8.01 Factoring Polynomials

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

44 [Challenging] • Target: The laws of exponents and scientific notation • Cross-Lesson: 9.01 Radicals & Exponents

A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

45 [Challenging] • Target: Patterns with rational and irrational numbers • Cross-Lesson: 1.01 Linear Equations & Lines

A data set recorded for patterns with rational and irrational numbers consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

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